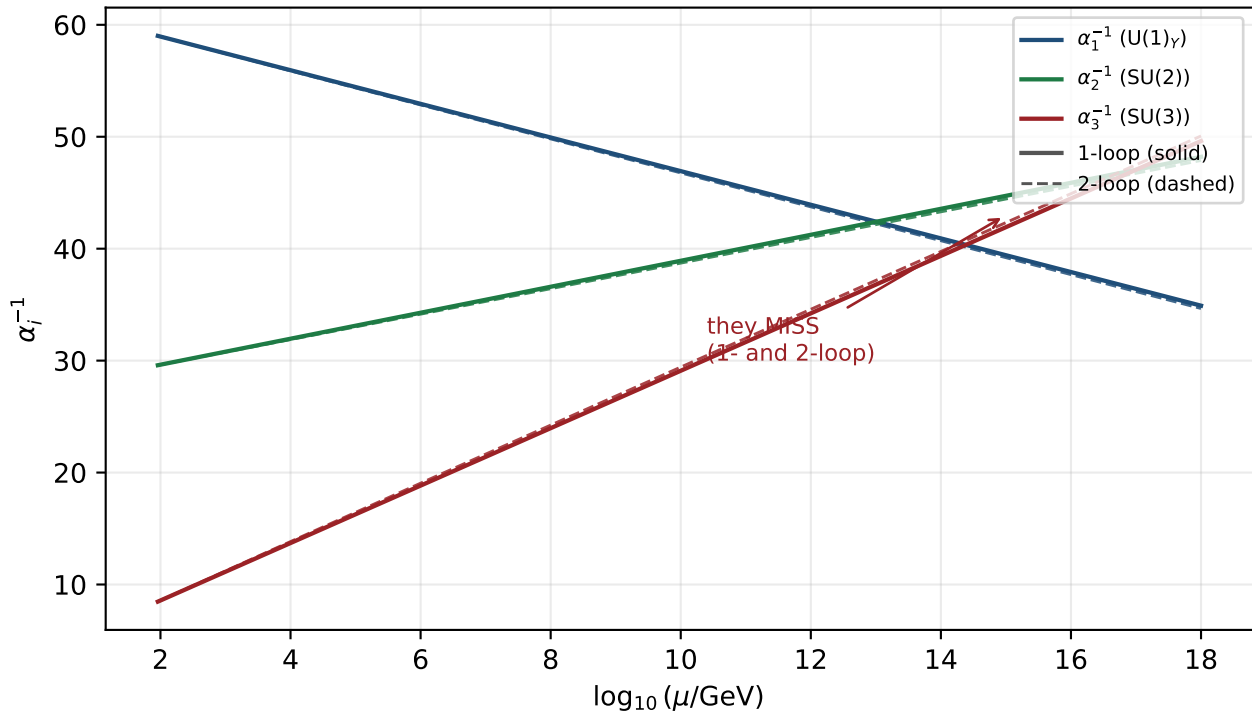


Spectral-action unification vs measured couplings (v246):
1- and 2-loop both MISS -- a tension, no rescue (SM content, no new state)



$g_1 = g_2 = g_3$, $\sin^2 \theta_W = 3/8$ is the spectral-action prediction; with SM content and no new states there is no admissible threshold source -- the 4d-GUT route is in tension (like NCG-SM), not closed (QFT4D.RGTEST.01).